# System Operation Overview

The millipede terrarium exhibit is made to be easy to use and requires little guidance from museum employees. A green LED ring light on the outside of the exhibit indicates that it is in an idle state under normal circumstances. The system starts its automated presentation cycle when a visitor presses the push-button input. While the internal lighting dims to improve the viewing experience, the external light changes from green to red to indicate occupancy. Concurrently, the speaker system starts playing the instructional narration, and the ultraviolet (UV) lighting system turns on, illuminating the fluorescent millipede display.

All systems immediately return to their default idle state after the programmed cycle is finished. The UV and audio systems go off, the interior lighting returns to normal brightness, and the external illumination changes back to green. Without the need for manual reset or intervention, this fully automatic loop guarantees that the display is always prepared for the next visitor.

# Routine Maintenance Guide

To ensure long-term reliability and performance, museum staff should perform simple routine maintenance checks on a scheduled basis:

## Daily Inspection

- Verify that the system turns on properly and shows the green idle light.
- To confirm full cycle operation (lighting, UV, and audio), press the button once.
- Examine the enclosure for any obvious damage or loose wires.

## Weekly Maintenance

- Examine all wired connections to make sure the parts are firmly fixed inside the engraved birch wood base.
- To keep the fluorescent display visible, keep the inside surfaces and viewing areas clean.
- Make sure that the LED strips and UV light are operating at maximum brightness.(8 bit PWM)

## Monthly Maintenance

- Check the clarity of the speaker and replace or modify it if the audio distortion gets worse.
- Verify the 3D-printed enclosure and casing's structural soundness.
- Verify that no electrical components are overheating or exhibiting unusual wear.

# Troubleshooting Guide

The following troubleshooting procedures should be used if the system does not function as intended:

## System Does Not Power On

- Make sure that the Arduino is getting power by checking the power supply connector.
- Look for jumper wires that are loose or disconnected.
- Verify that the power module is operating correctly.

## Button Press Does Not Start Cycle

- Examine the wiring for the push buttons and their connection to the Arduino input pin.
- Verify the program upload and make sure the right logic is operating.

## Lighting System Failure (LED or UV)

- Check the connections for the UV lamp and LED strip.
- Look for any broken parts or inadequate power.
- Make sure that the breadboard is receiving the appropriate voltage. 5V DC (or 3.3V DC if required by specific components)

## Audio Issues (Low Volume or Distortion)

- Examine the stability of the connections and speaker wire.
- Verify that the audio module is powered on correctly.
- If there are still problems with clarity, move or replace the speaker.

## System Freezing or Not Resetting

- Disconnect and reconnect the power to restart the system.
- Check for timing or loop issues in the Arduino code.
- Look for component stress or overheating.